BJT small-signal II

Answers and worked solutions · Day 12 · Wed 4 Nov · pack E7

5. Practice

From your course files

WS6 Q6 (course question).

Which DC method. $\beta R_E = 180(2.2\text{ k}) = 396$ kΩ < $10R_2 = 560$ kΩ, so use the exact method.

$R_{Th} = 220\text{ k} \parallel 56\text{ k} = 44.64$ kΩ, $E_{Th} = \dfrac{56}{276}(20) = 4.06$ V.

$$I_B = \frac{4.06 - 0.7}{44.64\text{ k} + 181(2.2\text{ k})} = 7.59\ \mu\text{A}, \qquad I_E = 181 I_B = 1.374\text{ mA}$$

1. $r_e = 26\text{ mV}/1.374\text{ mA} = \mathbf{18.92\ \Omega}$ (the official value; unrounded I_B = 7.58 µA gives 18.94 Ω).

2. $V_B = V_{BE} + I_E R_E = 0.7 + 1.374(2.2) = \mathbf{3.72\ V}$. V_C is not given in the solution: $V_C = 20 - I_C R_C = 20 - 1.365(4.7) = \mathbf{13.58\ V}$, so V_{CE} = 10.6 V (active).

3. $Z_i = R_{Th} \parallel \beta r_e = 44.64\text{ k} \parallel 180(18.92) = \mathbf{3.17\ k\Omega}$.

$r_o = 1/g_{os} = 1/30\ \mu\text{S} = 33.3$ kΩ. That is less than $10R_C$ = 47 kΩ, so keep it: $R_C \parallel r_o = 4.12$ kΩ.

$$A_V = -\frac{R_C \parallel r_o}{r_e} = -\frac{4.12\text{ k}}{18.92} = \mathbf{-217.8}$$

With unrounded r_e this is −217.4. Either is the same answer.

WS6 Q7 (course question).

This is a pnp common-base stage: the emitter goes through 6.8 kΩ to +6 V, the base is grounded and the collector goes through 4.7 kΩ to −10 V.

1. DC emitter loop:

$$I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6 - 0.7}{6.8\text{ k}\Omega} = 0.779\text{ mA}, \qquad r_e = \frac{26\text{ mV}}{0.779\text{ mA}} = \mathbf{33.38\ \Omega}$$

2. $Z_i = R_E \parallel r_e = 6.8\text{ k} \parallel 33.38 = \mathbf{33.22\ \Omega}$ and $Z_o = R_C = \mathbf{4.7\ k\Omega}$. ($r_o = 1/g_{os}$ = 100 kΩ ≥ $10R_C$, so it is ignored.)

3.

$$A_V \approx \frac{R_C}{r_e} = \frac{4.7\ \text{k}}{33.38} = \mathbf{140.8}$$

Positive: CB does not invert. Keeping α = 0.998 gives 140.5, so α hardly matters.

WS6 Q4 (course question).

$\beta = \infty$, so $\alpha = 1$. In the small-signal circuit the 10 kΩ goes from the collector to ground, the base is grounded, the current source is open, and v_i drives the emitter.

T model: r_e = 0.125 kΩ from base to emitter, αi_e from collector to base.

- Collector: $v_c = -\alpha i_e(10\ \text{k}\Omega)$
- Emitter: $i_e = -v_i/r_e$

$$\frac{v_c}{v_i} = \frac{\alpha(10\ \text{k})}{r_e} = \frac{1 \times 10\ \text{k}}{0.125\ \text{k}} = \mathbf{80}$$

A non-inverting (common-base) amplifier.

WS6 Q5 (course question).

1. DC sources off, capacitors shorted: v_{sig} and 2 kΩ drive the base; 10 kΩ from base to ground; r_π from base to emitter; the emitter is at AC ground through Z; $g_m v_{be}$ and r_o from collector to emitter; R_C = 8 kΩ and the 8 kΩ load both from collector to ground.

2.

$$v_y = -g_m v_{be}(r_o \parallel 8\text{k} \parallel 8\text{k}), \qquad v_{be} = v_{sig}\frac{r_\pi \parallel 10\text{k}}{(r_\pi \parallel 10\text{k}) + 2\text{k}}$$

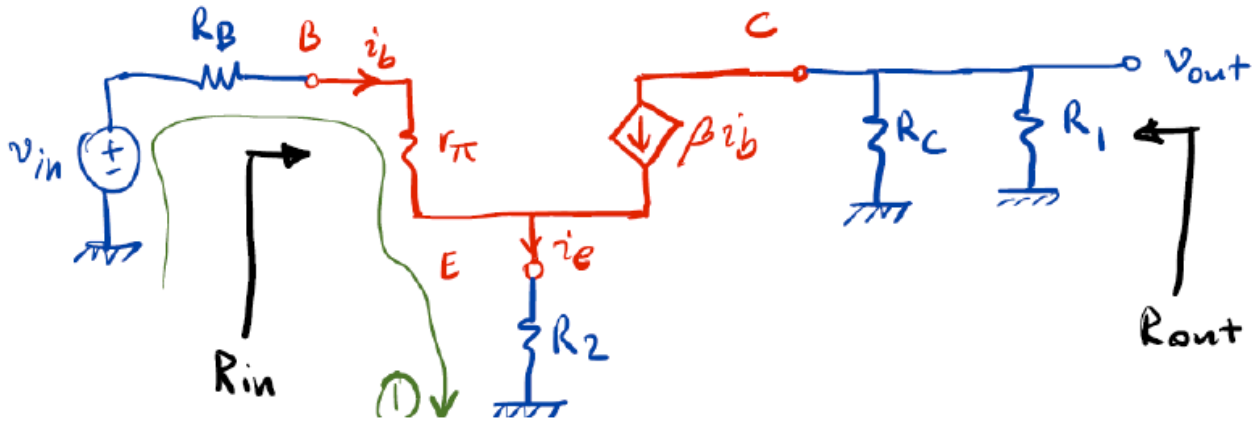
$r_\pi \parallel 10\text{k}$ = 2.016 kΩ (divider 0.502) and $r_o \parallel 8\text{k} \parallel 8\text{k}$ = 3.848 kΩ, so

$$\frac{v_y}{v_{sig}} = -39.604(0.502)(3.848) = \mathbf{-76.49\ V/V}$$

Inverting.

WS6 Q9 (course question).

Small-signal circuit. C_1 grounds the R_1–R_2 node, so R_1 ends up from the collector to ground (in parallel with R_C) and R_2 from the emitter to ground. The current source I_1 is open. The official drawing:



WS6 Q9: hybrid-π small-signal circuit from the official solution

Voltage gain.

$$v_{out} = -\beta i_b (R_C \parallel R_1) \qquad (1)$$

KVL around loop 1 with $i_e = (1+\beta) i_b$:

$$v_{in} = [R_B + r_\pi + (1+\beta) R_2] i_b \qquad (2)$$

$$\boxed{A_v = \frac{v_{out}}{v_{in}} = \frac{-\beta (R_C \parallel R_1)}{R_B + r_\pi + (1+\beta) R_2}}$$

Input impedance (looking into the base, which is how the solution defines it):

$$R_{in} = \frac{v_\pi + v_{R_2}}{i_b} = r_\pi + (1+\beta) R_2$$

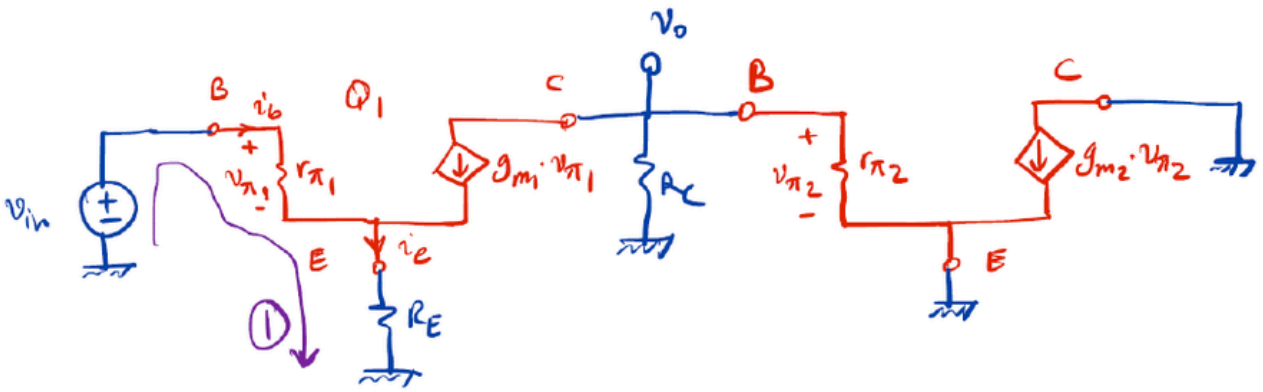
Seen from v_{in} it would be $R_B + r_\pi + (1+\beta) R_2$. Say which one you give.

Output impedance: with $v_{in} = 0$, $i_b = 0$, so the source is open and $R_{out} = R_C \parallel R_1$.

Check: with $R_2 = 0$ and $R_B = 0$ the gain collapses to $-g_m (R_C \parallel R_1)$, the bypassed CE result.

WS6 Q10 (course question).

Small-signal circuit (official drawing): Q_2's emitter is grounded, so its $r_{\pi 2}$ sits directly from Q_1's collector node to ground. Q_2's own collector goes to V_{CC} (AC ground) and does not affect v_{out}.



WS6 Q10: hybrid-π small-signal circuit from the official solution

The output node carries $R_C \parallel r_{\pi 2}$, and $g_{m1} v_{\pi 1}$ flows into it:

$$v_o = -g_{m1} v_{\pi 1} (R_C \parallel r_{\pi 2}) \qquad (1)$$

KVL in loop 1 with $i_e = (1+\beta) i_b$ and $i_b = v_{\pi 1} / r_{\pi 1}$:

$$v_{in} = [r_{\pi 1} + (1+\beta) R_E] \frac{v_{\pi 1}}{r_{\pi 1}} \qquad (2)$$

$$\boxed{\frac{v_o}{v_{in}} = \frac{-g_{m1} r_{\pi 1} (R_C \parallel r_{\pi 2})}{r_{\pi 1} + (1+\beta) R_E} = \frac{-(R_C \parallel r_{\pi 2})}{\dfrac{1}{g_{m1}} + \dfrac{1+\beta}{\beta} R_E}}$$

Here β is β_1; β_2 enters only through $r_{\pi 2}$.

WS6 Q11 (course question).

Why it fails. The speaker is connected directly from the collector of Q_1 to ground. At DC its solenoid is almost a short (under 1 Ω), so it pulls the collector to ground: $V_C \approx 0$. With $V_E = 0$, $V_{CE} \approx 0$, so Q_1 is driven into deep saturation and no longer works as an amplifier. The input coupling capacitor C_1 fixed the base bias, but the output is still DC-coupled.

Fix. Insert a coupling capacitor between the collector of Q_1 and the speaker. It blocks the DC component while passing the AC signal, so the speaker no longer disturbs the bias, and the microphone signal is still amplified.

New questions

N1 (easy) [5 marks].

a. Test: $\beta R_E = 150$ kΩ ≥ $10 R_2 = 120$ kΩ, so it is allowed. $V_B = \frac{12}{80}(15) = 2.25$ V, $V_E = 1.55$ V, $I_E = 1.55$ mA, $V_{CE} = 15 - 1.55(3.3 + 1) = 8.34$ V. Active. (The exact method gives 1.45 mA. The approximation is 7 % high, which is acceptable because the test passed.)

b. $r_e = 26/1.55 = 16.8\ \Omega$. $R' = 68\text{k} \parallel 12\text{k} = 10.2$ kΩ.
$Z_b = 150(16.8 + 1000) = 152.5$ kΩ. $Z_i = 10.2\text{k} \parallel 152.5\text{k} = \mathbf{9.56\ k\Omega}$.
$Z_o = R_C = \mathbf{3.3\ k\Omega}$. $A_v = -\dfrac{3300}{1016.8} = \mathbf{-3.25}$, against $-R_C/R_E = -3.30$ (1.7 % apart). Bypassed it would be $-3300/16.8 = -197$.

N2 (medium) [7 marks].

a. $I_E = (5 - 0.7)/2.2\text{k} = 1.955$ mA, $r_e = 26/1.955 = 13.30\ \Omega$,
$V_{CB} = 10 - 0.99(1.955)(3.9) = 2.45$ V > 0, so the CBJ is reverse biased (active).

b. $Z_i = R_E \parallel r_e = 2200 \parallel 13.30 = \mathbf{13.2\ \Omega}$. $Z_o = R_C = \mathbf{3.9\ k\Omega}$.
$A_{v_{NL}} = \alpha R_C/r_e = 0.99(3900)/13.30 = \mathbf{290}$.
$A_{v_L} = 0.99(3.9\text{k} \parallel 10\text{k})/13.30 = 0.99(2806)/13.30 = \mathbf{209}$. Both in phase.

c. $R_s = 50\ \Omega$: $A_{v_s} = \frac{13.2}{63.2}(209) = \mathbf{43.7}$. $R_s = 1$ kΩ: $A_{v_s} = \frac{13.2}{1013.2}(209) = \mathbf{2.72}$. The CB input is so low that a 1 kΩ source throws away 98.7 % of the signal. CB only makes sense when it is driven from a very low impedance (or from a current source, such as a CE collector in a cascode).

N3 (medium–hard, multi-part) [10 marks].

a. $A_{v_L} = -164 \times \frac{500}{500+3810} = \mathbf{-19.0}$. $A_{v_s} = \frac{3.60}{4.60}(-19.0) = \mathbf{-14.9}$. The 500 Ω load kills 88 % of the gain.

b. Emitter AC resistance: $R_E \parallel R_L = 3.3\text{k} \parallel 0.5\text{k} = 434.2\ \Omega$.
$Z_{b2} = 100(12.60 + 434.2) = 44.7$ kΩ, $Z_{i2} = 220\text{k} \parallel 44.7\text{k} = \mathbf{37.1\ k\Omega}$.

c. Stage 1 is loaded by Z_{i2}: $A_{v1} = -\frac{3810 \parallel 37.1\text{k}}{23.2} = -\frac{3455}{23.2} = \mathbf{-148.9}$. Follower: $A_{v2} = \frac{434.2}{434.2+12.60} = \mathbf{0.972}$. Overall: $A_{v_s} = \frac{3.60}{4.60}(-148.9)(0.972) = \mathbf{-113}$. That is 7.6 times better than (a).

d. The follower's source is the CE output, $Z_o = 3.81$ kΩ, in parallel with R_B:
$R'_s = 3.81\text{k} \parallel 220\text{k} = 3.745$ kΩ.
$Z_o = R_E \parallel \left(r_e + \frac{R'_s}{\beta}\right) = 3300 \parallel (12.60 + 37.45) = \mathbf{49.3\ \Omega}$. Using the slide's $R_E \parallel r_e = 12.6\ \Omega$ would be wrong by a factor of 4.

N4 (design) [6 marks].

a. Keep $R_{E1} + R_{E2} = 980\ \Omega$ so the DC loop, and therefore the Q-point, is unchanged. Put C_E across R_{E2} only. $\frac{R_C}{r_e + R_{E1}} = 10 \Rightarrow R_{E1} = 381 - 23.2 = 357.8\ \Omega$. Choose $R_{E1} = 360\ \Omega$ and $R_{E2} = 620\ \Omega$ (both E24, and they sum to exactly 980 Ω).

b. $A_v = -\frac{3810}{23.2+360} = \mathbf{-9.94}$.
$Z_i = 8.214\text{k} \parallel 275.5(383.2) = 8.214\text{k} \parallel 105.6\text{k} = \mathbf{7.62\ k\Omega}$. With β = 100: $8.214\text{k} \parallel 38.3\text{k} = \mathbf{6.76\ k\Omega}$.

c. r_e becomes $26/1.232 = 21.1\ \Omega$. Bypassed: $-164.1 \to -180.5$, a **10 %** rise. Your design: $-9.94 \to -10.00$, a **0.6 %** rise. The partial bypass makes the gain about 17 times less sensitive to bias drift.

N5 (hard) [8 marks].

a. At AC, R_C, R_F (at the output end) and R_L are in parallel:
$A_{v_L} = -\frac{3.9\text{k} \parallel 220\text{k} \parallel 3.9\text{k}}{13.08} = -\frac{1933}{13.08} = \mathbf{-147.7}$.

b. $\frac{R_F}{1-A_{v_L}} = \frac{220\text{k}}{148.7} = 1.479$ kΩ. $Z_i = \beta r_e \parallel 1.479\text{k} = 1570 \parallel 1479 = \mathbf{762\ \Omega}$. The load halves the gain, so the voltage across R_F is about half as large and R_F draws about half the current. It looks like a bigger resistor from the base, so Z_i rises.

c. $A_{v_s} = \frac{762}{762+600}(-147.7) = \mathbf{-82.6}$. A full nodal solution gives -82.63.

d. $\frac{507}{507+600}(-292.9)\frac{3900}{3900+3832} = -67.7$. That is 18 % low, because it uses the unloaded Z_i (507 Ω) when the real Z_i with R_L connected is 762 Ω. The two-port shortcut assumes Z_i does not depend on R_L. For a feedback stage it does.

6. Quick recall check

Question

1. $A_v \approx -R_C/(r_e + R_E) \approx -4.7/1.2 \approx -3.9$. β is missing (and nearly r_e too), so the gain is stable.
2. $Z_o = 2\text{k} \parallel (10 + 2000/100) = 2\text{k} \parallel 30\ \Omega \approx 29.6\ \Omega$.
3. The emitter follower (R_L is in the emitter and is reflected into Z_b) and collector feedback (Miller: Z_i depends on the gain).
4. $A_{v_s} = \dfrac{Z_i}{Z_i + R_s} A_{v_L}$.
5. DC: $R_{E1} + R_{E2}$. AC: R_{E1} only, $A_v \approx -R_C/(r_e + R_{E1})$.